THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 9	Pairs: $(360,156)\to(204,156)\to(156,48)\to(108,48)\to(60,48)\to(48,12)\to(36,12)\to(24,12)\to(12,12)\to(12,0)$.
2	**(d)** 4	$468=280\times1+188$; $280=188\times1+92$; $188=92\times2+4$; $92=4\times23+0$.
3	**(a)** 23	Multiples of 3: 16; of 5: 10; of both: 3.
4	**(d)** 18	The algorithm adds the digits.
5	**(c)** 100	LCM $=\dfrac{20\times25}{5}$.
6	**(d)** 19	$\lfloor\sqrt{373}\rfloor=19$.
7	**(a)** 7	$408=125\times3+33$; $125=33\times3+26$; $33=26\times1+7$; $26=7\times3+5$; $7=5\times1+2$; $5=2\times2+1$; $2=1\times2+0$.
8	**(b)** 4	$692=636\times1+56$; $636=56\times11+20$; $56=20\times2+16$; $20=16\times1+4$; $16=4\times4+0$.
9	**(d)** 6	Pairs: $(294,245)\to(245,49)\to(196,49)\to(147,49)\to(98,49)\to(49,49)\to(49,0)$.
10	**(d)** 4	$416=217\times1+199$; $217=199\times1+18$; $199=18\times11+1$; $18=1\times18+0$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$106=44\times2+18$; $44=18\times2+8$; $18=8\times2+2$; $8=2\times4+0$; gcd $=2$ (1); LCM $=\dfrac{106\times44}{2}=2332$ (1).	1+1
12	$\lfloor\sqrt{134}\rfloor=11$, so check $2,\dots,11$; The smallest divisor found is 2 since $134=2\times67$ (1); 134 is composite (1).	1+1
13	Replace (m,n) by $(n,m-n)$ re-ordered so the larger is first; $(224,154)\to(154,70)\to(84,70)\to(70,14)\to(56,14)\to(42,14)\to(28,14)\to(14,14)\to(14,0)$ (1); 8 subtractions, gcd $=14$ (1).	1+1
14	$120=60\times2+0$; gcd $=60$ (1); LCM $=\dfrac{60\times120}{60}=120$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Subtraction: $(805,690)\to(690,115)\to(575,115)\to(460,115)\to(345,115)\to(230,115)\to(115,115)\to(115,0)$, 7 steps (1); Division: $805=690\times1+115$; $690=115\times6+0$, 2 steps (1); gcd $=115$ and the division form is shorter (1).	1+1+1
16	Multiples of 3: 8; multiples of 5: 5 (1); Multiples of both (counted twice): 1 (1); Count $=8+5-1=12$ (1).	1+1+1
17	Rule: new $x=2x-4$ (1); Values: 23, 42, 80, 156, 308, 612 (1); Final $x=612$ (1).	1+1+1
	OR: Subtraction: $(330,300)\to(300,30)\to(270,30)\to(240,30)\to(210,30)\to(180,30)\to(150,30)\to(120,30)\to(90,30)\to(60,30)\to(30,30)\to(30,0)$, 11 steps (1); Division: $330=300\times1+30$; $300=30\times10+0$, 2 steps (1); gcd $=30$ and the division form is shorter (1).	1+1+1
18	Multiples of 3: 11; multiples of 5: 6 (1); Multiples of both (counted twice): 2 (1); Count $=11+6-2=15$ (1).	1+1+1
	OR: Rule: new $x=2x-3$ (1); Values: 20, 37, 71, 139 (1); Final $x=139$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Values of r: 2, 23, 234, 2341 (1); Final $r=2341$ (2); It reverses the digits (2).	1+2+2
	OR: Sequence: 9, 28, 14, 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (2); Number of steps $=19$ (3).	2+3
20	Values of r: 7, 71, 712, 7121 (1); Final $r=7121$ (2); It reverses the digits (2).	1+2+2
	OR: Sequence: 6, 3, 10, 5, 16, 8, 4, 2, 1 (2); Number of steps $=8$ (3).	2+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 10	Multiples of 3: 7; of 5: 4; of both: 1.
2	**(b)** 10	The algorithm adds the digits.
3	**(a)** 120	LCM $= \frac{40 \times 15}{5}$.
4	**(d)** 19	$\lfloor \sqrt{389} \rfloor = 19$.
5	**(b)** 5	$554 = 126 \times 4 + 50$; $126 = 50 \times 2 + 26$; $50 = 26 \times 1 + 24$; $26 = 24 \times 1 + 2$; $24 = 2 \times 12 + 0$.
6	**(d)** 3	$453 = 384 \times 1 + 69$; $384 = 69 \times 5 + 39$; $69 = 39 \times 1 + 30$; $39 = 30 \times 1 + 9$; $30 = 9 \times 3 + 3$; $9 = 3 \times 3 + 0$.
7	**(c)** 9	Pairs: $(280, 203) \to (203, 77) \to (126, 77) \to (77, 49) \to (49, 28) \to (28, 21) \to (21, 7) \to (14, 7) \to (7, 7) \to (7, 0)$.
8	**(b)** 7	$308 = 167 \times 1 + 141$; $167 = 141 \times 1 + 26$; $141 = 26 \times 5 + 11$; $26 = 11 \times 2 + 4$; $11 = 4 \times 2 + 3$; $4 = 3 \times 1 + 1$; $3 = 1 \times 3 + 0$.
9	**(c)** 21	Multiples of 3: 15; of 5: 9; of both: 3.
10	**(a)** 24	The algorithm adds the digits.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\lfloor \sqrt{133} \rfloor = 11$, so check $2, \ldots, 11$; The smallest divisor found is 7 since $133 = 7 \times 19$ (1); 133 is composite (1).	1+1
12	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first; $(186, 124) \to (124, 62) \to (62, 62) \to (62, 0)$ (1); 3 subtractions, gcd $= 62$ (1).	1+1
13	$111 = 72 \times 1 + 39$; $72 = 39 \times 1 + 33$; $39 = 33 \times 1 + 6$; $33 = 6 \times 5 + 3$; $6 = 3 \times 2 + 0$; gcd $= 3$ (1); LCM $= \frac{72 \times 111}{3} = 2664$ (1).	1+1
14	$\lfloor \sqrt{193} \rfloor = 13$, so check $2, \ldots, 13$; No number in this range divides it (1); 193 is prime (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Subtraction: $(874, 368) \to (506, 368) \to (368, 138) \to (230, 138) \to (138, 92) \to (92, 46) \to (46, 46) \to (46, 0)$, 7 steps (1); Division: $874 = 368 \times 2 + 138$; $368 = 138 \times 2 + 92$; $138 = 92 \times 1 + 46$; $92 = 46 \times 2 + 0$, 4 steps (1); gcd $= 46$ and the division form is shorter (1).	1+1+1
16	Multiples of 3: 7; multiples of 5: 4 (1); Multiples of both (counted twice): 1 (1); Count $= 7 + 4 - 1 = 10$ (1).	1+1+1
17	Rule: new $x = 2x - 1$ (1); Values: 28, 55, 109, 217, 433, 865 (1); Final $x = 865$ (1).	1+1+1
	OR: Subtraction: $(855, 303) \to (552, 303) \to (303, 249) \to (249, 54) \to (195, 54) \to (141, 54) \to (87, 54) \to (54, 33) \to (33, 21) \to (21, 12) \to (12, 9) \to (9, 3) \to (6, 3) \to (3, 3) \to (3, 0)$, 14 steps (1); Division: $855 = 303 \times 2 + 249$; $303 = 249 \times 1 + 54$; $249 = 54 \times 4 + 33$; $54 = 33 \times 1 + 21$; $33 = 21 \times 1 + 12$; $21 = 12 \times 1 + 9$; $12 = 9 \times 1 + 3$; $9 = 3 \times 3 + 0$, 8 steps (1); gcd $= 3$ and the division form is shorter (1).	1+1+1
18	Multiples of 3: 8; multiples of 5: 5 (1); Multiples of both (counted twice): 1 (1); Count $= 8 + 5 - 1 = 12$ (1).	1+1+1
	OR: Rule: new $x = 2x - 3$ (1); Values: 12, 21, 39, 75 (1); Final $x = 75$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Values of r: 6, 61, 612, 6126 (1); Final $r = 6126$ (2); It reverses the digits (2).	1+2+2
	OR: Sequence: 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (2); Number of steps $= 14$ (3).	2+3
20	Values of r: 5, 59, 596, 5964 (1); Final $r = 5964$ (2); It reverses the digits (2).	1+2+2
	OR: Sequence: 12, 6, 3, 10, 5, 16, 8, 4, 2, 1 (2); Number of steps $= 9$ (3).	2+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **World of Algorithms** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 702	LCM $= \frac{26 \times 54}{2}$.
2	**(c)** 8	$\lfloor \sqrt{67} \rfloor = 8$.
3	**(d)** 4	$493 = 156 \times 3 + 25$; $156 = 25 \times 6 + 6$; $25 = 6 \times 4 + 1$; $6 = 1 \times 6 + 0$.
4	**(c)** 2	$656 = 562 \times 1 + 94$; $562 = 94 \times 5 + 92$; $94 = 92 \times 1 + 2$; $92 = 2 \times 46 + 0$.
5	**(a)** 5	Pairs: $(357, 255) \to (255, 102) \to (153, 102) \to (102, 51) \to (51, 51) \to (51, 0)$.
6	**(b)** 4	$391 = 262 \times 1 + 129$; $262 = 129 \times 2 + 4$; $129 = 4 \times 32 + 1$; $4 = 1 \times 4 + 0$.
7	**(d)** 24	Multiples of 3: 17; of 5: 10; of both: 3.
8	**(d)** 16	The algorithm adds the digits.
9	**(c)** 60	LCM $= \frac{60 \times 15}{15}$.
10	**(b)** 13	$\lfloor \sqrt{193} \rfloor = 13$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first; $(228, 152) \to (152, 76) \to (76, 76) \to (76, 0)$ (1); 3 subtractions, gcd $= 76$ (1).	1+1
12	$96 = 90 \times 1 + 6$; $90 = 6 \times 15 + 0$; gcd $= 6$ (1); LCM $= \frac{96 \times 90}{6} = 1440$ (1).	1+1
13	$\lfloor \sqrt{110} \rfloor = 10$, so check $2, \ldots, 10$; The smallest divisor found is 2 since $110 = 2 \times 55$ (1); 110 is composite (1).	1+1
14	Replace (m, n) by $(n, m - n)$ re-ordered so the larger is first; $(217, 186) \to (186, 31) \to (155, 31) \to (124, 31) \to (93, 31) \to (62, 31) \to (31, 31) \to (31, 0)$ (1); 7 subtractions, gcd $= 31$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Subtraction: $(511, 392) \to (392, 119) \to (273, 119) \to (154, 119) \to (119, 35) \to (84, 35) \to (49, 35) \to (35, 14) \to (21, 14) \to (14, 7) \to (7, 7) \to (7, 0)$, 11 steps (1); Division: $511 = 392 \times 1 + 119$; $392 = 119 \times 3 + 35$; $119 = 35 \times 3 + 14$; $35 = 14 \times 2 + 7$; $14 = 7 \times 2 + 0$, 5 steps (1); gcd $= 7$ and the division form is shorter (1).	1+1+1
16	Multiples of 3: 6; multiples of 5: 4 (1); Multiples of both (counted twice): 1 (1); Count $= 6 + 4 - 1 = 9$ (1).	1+1+1
17	Rule: new $x = 2x - 3$ (1); Values: 14, 25, 47, 91, 179, 355 (1); Final $x = 355$ (1).	1+1+1
	OR: Subtraction: $(825, 477) \to (477, 348) \to (348, 129) \to (219, 129) \to (129, 90) \to (90, 39) \to (51, 39) \to (39, 12) \to (27, 12) \to (15, 12) \to (12, 3) \to (9, 3) \to (6, 3) \to (3, 3) \to (3, 0)$, 14 steps (1); Division: $825 = 477 \times 1 + 348$; $477 = 348 \times 1 + 129$; $348 = 129 \times 2 + 90$; $129 = 90 \times 1 + 39$; $90 = 39 \times 2 + 12$; $39 = 12 \times 3 + 3$; $12 = 3 \times 4 + 0$, 7 steps (1); gcd $= 3$ and the division form is shorter (1).	1+1+1
18	Multiples of 3: 13; multiples of 5: 8 (1); Multiples of both (counted twice): 2 (1); Count $= 13 + 8 - 2 = 19$ (1).	1+1+1
	OR: Rule: new $x = 2x - 1$ (1); Values: 13, 25, 49, 97, 193, 385 (1); Final $x = 385$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Values of r: 4, 49, 490, 4905 (1); Final $r = 4905$ (2); It reverses the digits (2).	1+2+2
	OR: Sequence: 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 (2); Number of steps $= 16$ (3).	2+3
20	Values of r: 5, 57, 571, 5711 (1); Final $r = 5711$ (2); It reverses the digits (2).	1+2+2
	OR: Sequence: 10, 5, 16, 8, 4, 2, 1 (2); Number of steps $= 6$ (3).	2+3

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